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United States Patent	12404135
Kind Code	B2
Date of Patent	September 02, 2025
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Carrier tape processing device and carrier tape processing method

Abstract

A roll body taking-out unit that takes out a roll body from a reel that holds a carrier tape storing components in a state of the roll body and stores the roll body in a case is provided. The roll body taking-out unit includes a separation unit that separates the roll body from the reel, and a roll body storage unit that stores the roll body separated from the reel by the separation unit in a case. Further, the roll body storage unit includes a case holding part that holds an empty case, and a chute part that stores the roll body separated from the reel by the separation unit one by one in the empty case held by the case holding part.

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Appl. No.:	17/904858
Filed (or PCT Filed):	January 21, 2021
PCT No.:	PCT/JP2021/001952
PCT Pub. No.:	WO2021/192557
PCT Pub. Date:	September 30, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230111399 A1	Apr. 13, 2023

Foreign Application Priority Data

Publication Classification

Int. Cl.: **B65H19/22** (20060101); **B65H19/30** (20060101); **H05K13/02** (20060101); **H05K13/04** (20060101)

U.S. Cl.:

CPC **B65H19/2292** (20130101); **B65H19/30** (20130101); **H05K13/021** (20130101); B65H2301/41445 (20130101); B65H2301/4185 (20130101); B65H2301/418523 (20130101); B65H2301/418526 (20130101); B65H2301/41854 (20130101); B65H2301/41858 (20130101); B65H2301/419 (20130101); B65H2701/11332 (20130101); B65H2701/378 (20130101); B65H2701/50 (20130101); B65H2701/51 (20130101); H05K13/0419 (20180801)

Field of Classification Search

CPC: H05K (13/02); H05K (13/021); H05K (13/0084); H05K (13/086); H05K (13/0419); H05K (13/0434); B65H (19/2292); B65H (19/30); B65H (2301/4186); B65H (2301/419); B65H (2301/41445); B65H (2301/41446); B65H (2301/4185); B65H (2301/418523); B65H (2301/41854); B65H (2301/41856); B65H (2301/41858); B65H (2701/11332); B65H (2701/37); B65H (27/378); B65H (2701/50); B65H (2701/51); B65D (85/00); B65D (85/66); B65D (85/67); B65D (85/671); B65D (85/672)

USPC: 242/473.5; 242/474

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a carrier tape processing device and a carrier tape processing method for processing a carrier tape storing components.

BACKGROUND ART

(2) Conventionally, as a component supply unit in a component mounting device in which a component is mounted on a substrate, a tape feeder that supplies a component to a component

extraction position by conveying a carrier tape storing the component is known. The carrier tape used in the tape feeder is wound around a reel as a holding body thereof, and transportation and storage of the carrier tape, setting of the carrier tape with respect to the component supply unit, and the like are performed in a state of being wound around the reel (in a state of being a roll body with reel).

(3) In such a tape feeder, the reel is a standard product, and it is difficult to reduce a size in a width direction of the reel, and an empty reel is generated after the use of the carrier tape is finished, and it takes time and effort to process and store the reel. For this reason, there has been proposed a device capable of supplying a carrier tape to a component supply unit in a state of a roll body in which a reel is eliminated and the carrier tape is wound in a roll shape (for example, PTL 1 below). In PTL 1, a roll body of a carrier tape is put into a storage part of a component supply unit from means for refilling components (component supply means), and an unnecessary reel does not remain at a mounting work site after the component supply unit finishes pulling out the carrier tape.

CITATION LIST

Patent Literature

(4) PTL 1: Unexamined Japanese Patent Publication No. H5-21990

SUMMARY OF THE INVENTION

(5) Therefore, an object of the present disclosure is to provide a carrier tape processing device and a carrier tape processing method capable of realizing a component supply form without a reel with an inexpensive configuration.

(6) A carrier tape processing device of the present disclosure includes a roll body taking-out unit that takes out a roll body from a holding body that holds a carrier tape storing components in a state of the roll body and stores the roll body in a case.

(7) A carrier tape processing method of the present disclosure includes a roll body taking-out and storing step of taking out a roll body from a holding body that holds a carrier tape storing components in a state of the roll body and storing the roll body in a case.

(8) A carrier tape processing device of the present disclosure includes a roll body taking-out unit that takes out a roll body from a holding body that holds a carrier tape storing components in a state of the roll body and stores the roll body in a case including a storage unit, a component information acquisition unit that acquires component information that is information on components, and a writing unit that writes the component information acquired by the component information acquisition unit in the storage unit.

(9) A carrier tape processing method of the present disclosure includes a roll body taking-out and storing step of taking out a roll body from a holding body that holds a carrier tape storing components in a state of the roll body and storing the roll body in a case including a storage unit, and a writing step of writing component information that is information on components, in the storage unit.

(10) According to the present disclosure, a component supply form without a reel can be realized with an inexpensive configuration.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic configuration diagram of a component mounting system according to an exemplary embodiment of the present disclosure.

(2) FIG. 2 is a side view of a component mounting device included in the component mounting system according to the exemplary embodiment of the present disclosure.

(3) FIG. 3 is a perspective view of a roll body in a case used in the component mounting device according to the exemplary embodiment of the present disclosure.

- (4) FIG. 4 is a perspective view of a roll body with a reel according to the exemplary embodiment of the present disclosure.
- (5) FIG. 5 is an exploded view of a reel of the roll body with a reel used when a carrier tape processing device according to the exemplary embodiment of the present disclosure manufactures the roll body in a case.
- (6) FIG. 6A is a perspective view of a case used when the carrier tape processing device according to the exemplary embodiment of the present disclosure manufactures a roll body in a case.
- (7) FIG. 6B is a partially enlarged view of a case used when the carrier tape processing device according to the exemplary embodiment of the present disclosure manufactures the roll body in a case.
- (8) FIG. 7A is a perspective view of a roll body in a case manufactured by the carrier tape processing device according to the exemplary embodiment of the present disclosure.
- (9) FIG. 7B is a partially enlarged view of the roll body in a case manufactured by the carrier tape processing device according to the exemplary embodiment of the present disclosure.
- (10) FIG. 8A is a diagram illustrating a usage mode of the roll body in a case manufactured by the carrier tape processing device according to the exemplary embodiment of the present disclosure.
- (11) FIG. 8B is a diagram illustrating a usage mode of the roll body in a case manufactured by the carrier tape processing device according to the exemplary embodiment of the present disclosure.
- (12) FIG. 8C is a diagram illustrating a usage mode of the roll body in a case manufactured by the carrier tape processing device according to the exemplary embodiment of the present disclosure.
- (13) FIG. 9 is a perspective view of the carrier tape processing device according to the exemplary embodiment of the present disclosure.
- (14) FIG. 10 is a side view of the carrier tape processing device according to the exemplary embodiment of the present disclosure.
- (15) FIG. 11 is a front view of a main part of the carrier tape processing device according to the exemplary embodiment of the present disclosure.
- (16) FIG. 12A is a diagram for describing an operation of disassembling the reel by the carrier tape processing device according to the exemplary embodiment of the present disclosure.
- (17) FIG. 12B is a diagram for describing an operation of disassembling the reel by the carrier tape processing device according to the exemplary embodiment of the present disclosure.
- (18) FIG. 12C is a diagram for describing an operation of disassembling the reel by the carrier tape processing device according to the exemplary embodiment of the present disclosure.
- (19) FIG. 13 is a diagram illustrating a state immediately after the reel is disassembled by the carrier tape processing device according to the exemplary embodiment of the present disclosure.
- (20) FIG. 14 is a perspective view of a storage warehouse according to the exemplary embodiment of the present disclosure.
- (21) FIG. 15 is a block diagram illustrating a control system of the component mounting system according to the exemplary embodiment of the present disclosure.
- (22) FIG. 16 is a flowchart illustrating a flow of work of manufacturing the roll body in a case by the carrier tape processing device according to the exemplary embodiment of the present disclosure.
- (23) FIG. 17 is a flowchart illustrating a flow of work of storing the roll body in a case in the storage warehouse according to the exemplary embodiment of the present disclosure.
- (24) FIG. 18 is a diagram illustrating an image of an operation of the roll body in a case according to the exemplary embodiment of the present disclosure.

DESCRIPTION OF EMBODIMENT

- (25) Before describing exemplary embodiments of the present disclosure, a problem in a conventional device will be briefly described.
- (26) The device described in PTL 1 requires a large-scale device (component supply unit) for supplying the roll body of the carrier tape to the component supply unit, and has a problem of cost.

(27) Hereinafter, an exemplary embodiment of the present disclosure will be described with reference to the drawings. FIG. 1 is a configuration diagram of component mounting system 1 including a carrier tape processing device according to an exemplary embodiment of the present disclosure. Component mounting system 1 includes manufacturing line 2, carrier tape processing device 3 related to supply of components supplied to manufacturing line 2, and storage warehouse 4 related to storage and management of components. Manufacturing line 2 manufactures mounting substrate JK in which components are mounted on substrate KB by performing work while transferring substrate KB between a plurality of devices connected in series.

(28) In FIG. 1, manufacturing line 2 is connected to management computer 6 through information management terminal 5. Management computer 6 manages an operation of each device constituting manufacturing line 2 from management computer 6. Further, as illustrated in FIG. 1, carrier tape processing device 3 and storage warehouse 4 are also connected to management computer 6. Management computer 6 also manages operations of carrier tape processing device 3 and storage warehouse 4. As illustrated in FIG. 1, worker terminal 7 is connected to management computer 6, and a worker of component mounting system 1 can perform various operation inputs from worker terminal 7 to component mounting system 1.

(29) First, manufacturing line 2 will be described. In FIG. 1, manufacturing line 2 includes substrate supply device 11, printing device 12, post-printing inspection device 13, a plurality of component mounting devices 14, post-mounting inspection device 15, reflow device 16, final inspection device 17, and substrate recovery device 18.

(30) Substrate supply device 11 sequentially supplies substrate KB to printing device 12 on a downstream side. Printing device 12 carries in substrate KB supplied from substrate supply device 11, applies paste solder to an electrode formed on a surface of substrate KB, and carries out substrate KB to post-printing inspection device 13 on the downstream side. Post-printing inspection device 13 carries in substrate KB carried out from printing device 12, and inspects substrate KB by observing with a camera whether or not there is a place with a defective solder application state, and then carries out substrate KB to component mounting devices 14 on the downstream side.

(31) Each of component mounting devices 14 mounts the component on substrate KB carried in from an upstream side and carries out the component to the downstream side. Component mounting device 14 located on the most downstream side carries out substrate KB to post-mounting inspection device 15 located on the downstream side. Component mounting device 14 will be described later.

(32) Post-mounting inspection device 15 carries in substrate KB carried out from component mounting device 14 located on the most downstream side, inspects substrate KB by observing with a camera whether or not there is a place where the component is mounted in a defective state, and then carries out substrate KB to reflow device 16 on the downstream side. Reflow device 16 carries in substrate KB carried out from post-mounting inspection device 15, and causes substrate KB to pass through a reflow furnace to melt and solidify the solder, thereby bonding the component to the electrode. Final inspection device 17 carries in substrate KB that has passed through reflow device 16, observes the bonding state of the component to the electrode with a camera to inspect the bonding state, and then carries out substrate KB to substrate recovery device 18 on the downstream side. Substrate recovery device 18 receives and recovers substrate KB carried out from final inspecting device 17.

(33) Next, component mounting device 14 will be described with reference to FIG. 2. In FIG. 2, base cover 22 is provided on base 21 of component mounting device 14, and substrate transport path 24 that conveys substrate KB in a horizontal direction is provided in working space 23 between base 21 and base cover 22.

(34) Feeder carriages 25 are connected to positions on both sides of substrate transport path 24 on base 21. A plurality of component supply units 26 are attached to each of feeder carriages 25. Here, component supply unit 26 is a tape feeder, and supplies components BH one by one to a

predetermined component supply position by conveying carrier tape 27 by sprocket 26S.

(35) In FIG. 2, mounting head 32 that is moved in a horizontal plane direction by head moving mechanism 31 is provided in working space 23. Component suction nozzle 33 is provided to extend downward in mounting head 32. Component BH supplied by component supply unit 26 is sucked to a lower end of component suction nozzle 33.

(36) When substrate transport path 24 carries in substrate KB from the upstream side and positions substrate KB, each component mounting device 14 causes mounting head 32 to repeatedly perform the mounting turn while causing component supply unit 26 to supply component BH. In one mounting turn, mounting head 32 performs an operation of sucking and picking up component BH supplied by component supply unit 26 and an operation of mounting component BH at a predetermined component mounting position on substrate KB in this order. When all components BH to be mounted on substrate KB are mounted by repeatedly performing the mounting turns on mounting head 32, substrate transport path 24 is operated to carry out substrate KB to the downstream side.

(37) Here, in the present exemplary embodiment, carrier tape 27 used by component mounting device 14 for supplying component BH is fed out from roll body 41 in a case. Here, “roll body 41 in a case” refers to roll body 27R in which carrier tape 27 is rolled and stored in case 42 (see also FIG. 3). Roll body 41 in a case is different from roll body 43 with a reel (FIG. 4) that is conventionally used. Here, “roll body 43 with a reel” refers to roll body 27R held by reel 44.

(38) In the present exemplary embodiment, reel 44 of roll body 43 with a reel is assumed to have two side plates 45 and winding core 46 disposed between two side plates 45 as illustrated in FIG. 5. In the present exemplary embodiment, it is assumed that winding core 46 is coupled to one of side plates 45. It is assumed that a plurality of recesses 46K provided in winding core 46 of one of side plates 45 and a plurality of protrusions 45T provided on a side of the other of side plates 45 are fitted to each other to connect two side plates 45. Therefore, when reel 44 is disassembled in the present exemplary embodiment, reel 44 is divided into two parts, a part including one of side plates 45 and a part including the other of side plates 45.

(39) Here, case 42 of roll body 41 in a case will be described. As illustrated in FIG. 6A, case 42 includes left and right side walls 51, bottom wall 52, front wall 53, and rear wall 54, and includes upper opening 55 and rear opening 56. Radio tag 42M is attached to an outer surface of front wall 53. Two upper and lower projection bases 51B projecting inward from an inner surface of one of side walls 51 are provided on an upper part of rear wall 54, and two projection bases 51B are provided with tape tip end holding parts 42K projecting outward in the horizontal direction (FIG. 6B).

(40) As illustrated in FIGS. 7A and 7B, when roll body 41 in a case is stored or transported, tape tip end holding part 42K provided in case 42 is inserted into feed hole 27K (hole engaged with an outer peripheral teeth of sprocket 26S) of carrier tape 27 to be locked. Note that the case where roll body 41 in a case is stored or transported refers to a case where roll body 27R is stored in case 42, and a case where carrier tape 27 is not pulled out from roll body 27R to be used. In the present exemplary embodiment, two tape tip end holding parts 42K provided in case 42 are inserted into and locked to two feed holes 27K. This prevents carrier tape 27 from coming off from case 42 of roll body 41 in a case during storage or transportation, and thus prevents entire roll body 27R from falling off from case 42.

(41) On the other hand, when carrier tape 27 is pulled out from roll body 41 in a case for use, two tape tip end holding parts 42K are removed from feed hole 27K of carrier tape 27, and furthermore, front wall 53 is brought into an attitude to be a lower surface as illustrated in FIG. 3. Carrier tape 27 can be fed to component supply unit 26 by pulling out carrier tape 27 from rear opening 56 in such an attitude that front wall 53 is a lower surface. As described above, by providing tape tip end holding part 42K in case 42, a position of a tip end part of carrier tape 27 in case 42 is unified, so that the operation of taking out the tip end part of carrier tape 27 from case 42 can be easily

performed. Further, it is possible to easily cope with a case where the work of taking out the tip end part of carrier tape **27** is performed by an automated facility such as a robot.

(42) As a usage mode of roll body **41** in a case with respect to component supply unit **26**, for example, there is a mode of being attached to feeder carriage **25** as in roll body **41** in a case illustrated on a left side of FIG. **2**, and a mode of being attached to component supply unit **26** via attachment **26A** as in roll body **41** in a case illustrated on a right side of FIG. **2** and FIG. **8A**. Alternatively, as illustrated in FIG. **8B**, entire roll body **41** in a case may be stored inside component supply unit **26**. Alternatively, as illustrated in FIG. **8C**, there is also a mode in which roll body **27R** is taken out from roll body **41** in a case, and only roll body **27R** is stored inside component supply unit **26**.

(43) As described above, in the present exemplary embodiment, case **42** of roll body **41** in a case can have a role of a support means of roll body **27R** when roll body **27R** is installed outside component supply unit **26** to be used (a role of reel **44** in roll body **43** with a reel).

(44) Next, carrier tape processing device **3** will be described. Carrier tape processing device **3** is a device that processes carrier tape **27** used in component mounting device **14** to manufacture roll body **41** in a case. Carrier tape processing device **3** is configured to take out roll body **27R** from reel **44** of conventional roll body **43** with a reel, store roll body **27R** in case **42**, and recover reel **44**.

(45) As illustrated in FIGS. **9**, **10**, and **11**, carrier tape processing device **3** according to the present exemplary embodiment includes reel supply unit **61**, reel conveyance unit **62**, roll body taking-out unit **63**, recovery unit **64**, reading unit **65**, and writing unit **66**.

(46) Reel supply unit **61** has a function of supplying roll body **43** with a reel. As illustrated in FIGS. **9** and **10**, reel supply unit **61** includes, for example, a container-shaped member opened upward, and holds a plurality of roll bodies **43** with reels in a state of being arranged in a lateral direction.

(47) Reel conveyance unit **62** has a function of gripping roll body **43** with a reel supplied by reel supply unit **61** and moving the roll body with a reel upward from roll body taking-out unit **63**. As illustrated in FIGS. **9** and **10**, reel conveyance unit **62** includes gripping head **71** and gripping head moving mechanism **72**. Gripping head **71** grips roll body **43** with a reel supplied by reel supply unit **61** by two gripping parts **71H** extending downward. Gripping head moving mechanism **72** moves gripping head **71** gripping roll body **43** with a reel in the horizontal plane direction and the vertical direction.

(48) Roll body taking-out unit **63** has a function of taking out roll body **27R** from reel **44** of roll body **43** with a reel, and storing the taken-out roll body in case **42**. Roll body taking-out unit **63** includes separation unit **63A** that separates roll body **27R** from reel **44** of roll body **43** with a reel, and roll body storage unit **63B** that stores roll body **27R** separated from reel **44** by separation unit **63A** in case **42**.

(49) As illustrated in FIGS. **9** and **10**, separation unit **63A** includes separator **73** and separator driver **74**. Separator **73** includes a pair of movable plates **73S** facing each other in the horizontal direction and extending in the vertical direction. Each of movable plates **73S** is provided with U-shaped escape part **73N** recessed downward from an upper edge part (FIG. **10**).

(50) In FIGS. **9** and **10**, separator driver **74** includes a pair of (two) arm parts **74A** and arm driver **74B** that moves two arm parts **74A**. One end parts of two arm parts **74A** are attached to two movable plates **73S**, respectively, and extend in the substantially horizontal direction. Arm driver **74B** can change an interval between two arm parts **74A** (and thus separator **73**) by moving two arm parts **74A** in the horizontal direction.

(51) In FIGS. **9** and **10**, roll body storage unit **63B** includes case holding part **63Ba** and chute part **63Bb**. Roll body storage unit **63B** has a function of holding empty case **42** and conveying the same in the horizontal direction. Chute part **63Bb** has a function of guiding roll body **27R** such that roll body **27R** separated from reel **44** by separation unit **63A** is stored one by one in empty case **42** held by case holding part **63Ba**.

(52) As illustrated in FIGS. **9**, **10**, and **11**, case holding part **63Ba** includes a conveyor mechanism extending in the horizontal direction. Case holding part **63Ba** holds each of the plurality of cases **42** in a state of being arranged in a line such that upper opening **55** faces upward, and intermittently conveys the same in the horizontal direction.

(53) As illustrated in FIGS. **9**, **10**, and **11**, chute part **63Bb** includes a pair of (two) chute members **63T**. Two chute members **63T** are arranged to face each other in a direction in which two movable plates **73S** constituting separator **73** face each other. Two chute members **63T** are disposed below separator **73**.

(54) In FIG. **11**, each of two chute members **63T** has a shape in which upper end part **63J** expands upward. An interval between the upper end parts of two chute members **63T** is wider than an interval between the lower ends of two movable plates **73S** constituting separator **73**. Further, an interval between two chute members **63T** becomes narrower toward the lower end side (FIG. **11**).

(55) Chute part **63Bb** guides roll body **27R** separated from reel **44** by reel **44** being disassembled by separation unit **63A** downward, thereby storing roll body **27R** in case **42** located below the chute part. Accordingly, one roll body **27R** is stored in one case **42**.

(56) Recovery unit **64** has a function of recovering reel **44** disassembled by separation unit **63A**. As illustrated in FIGS. **9** and **10**, recovery unit **64** includes a pair of reel discharge chutes **75** and recovery box **76**.

(57) Two reel discharge chutes **75** have a tubular shape in which an opening part becomes wider toward an upper side. Upper end opening **75K** of each of two reel discharge chutes **75** is located on the side of separator **73** (the side of each of two movable plates **73S**). Recovery box **76** is a container-shaped member opened upward, and is provided below two reel discharge chutes **75**.

(58) When reel **44** is disassembled by separation unit **63A**, first, roll body **43** with a reel gripped by gripping head **71** of reel conveyance unit **62** is positioned above separator **73** (FIG. **11**). Then, after the interval between the upper end parts of two movable plates **73S** constituting separator **73** is set to be smaller than the interval between two side plates **45** constituting reel **44** of roll body **43** with a reel (arrow A in FIG. **12A**), gripping head **71** is lowered relative to separator **73** so that the upper end parts of two movable plates **73S** are inserted between two side plates **45** from below (arrow B in FIG. **12A**).

(59) When the upper end part of each of two movable plates **73S** is inserted between two side plates **45** from below, arm driver **74B** operates to increase the interval between two movable plates **73S**. At this time, the interval between two movable plates **73S** is set to be larger than the interval between two side plates **45** (arrow C illustrated in FIG. **12B**). As a result, movable plate **73S** pushes and spreads side plate **45** outward to widen a gap between side plate **45** and roll body **27R**.

(60) When the interval between side plate **45** and roll body **27R** increases by increasing the interval between two movable plates **73S** (FIG. **12B**), gripping head **71** is further lowered relative to separator **73** (arrow D illustrated in FIG. **12C**). Thus, each of two movable plates **73S** is inserted into the gap between widened side plate **45** and roll body **27R**. At this time, winding core **46** enters escape part **73N** formed in each of movable plates **73S**.

(61) When each of two movable plates **73S** is inserted between side plate **45** and roll body **27R**, arm driver **74B** further widens the interval between two movable plates **73S** (arrow E illustrated in FIG. **13**). As a result, two side plates **45** are pulled off, and the plurality of recesses **46K** provided on winding core **46** of one of side plates **45** and the plurality of protrusions **45T** provided on the other of side plates **45** are separated (FIG. **13**). As a result, reel **44** is divided into two parts, that is, a part including one of side plates **45** and a part including the other of side plates **45**. Roll body **27R** separated from reel **44** passes between two chute parts **63Bb** constituting roll body storage unit **63B** and is stored in empty case **42** located immediately below roll body storage unit **63B**.

(62) On the other hand, side plate **45** on one side of two divided reels **44** falls into upper end opening **75K** of reel discharge chute **75** on one side, and side plate **45** on the other side falls into upper end opening **75K** of reel discharge chute **75** on the other side (FIG. **13**). Each of these two

side plates **45** falls through reel discharge chute **75**, enters recovery box **76**, and is recovered.

(63) In FIG. **9**, reading unit **65** includes a reader or a camera that optically reads an identifier (symbol) such as a barcode or a two-dimensional code printed (recorded) on the outer surface of reel **44**, and is provided in the vicinity of reel supply unit **61**. The identifier includes component information of component BH stored in carrier tape **27**. In the present exemplary embodiment, reading unit **65** reads the code printed on code label **43L** (FIGS. **9** and **10**) attached to reel **44** of roll body **43** with a reel held by reel supply unit **61** to acquire (read) the component information. Broken line Br in FIGS. **9** and **10** schematically indicates a reading range of reading unit **65**. Here, the “component information” is information regarding component BH stored in carrier tape **27** of roll body **43** with a reel, and may include at least one of a type of component BH, a component name, characteristics, a date of manufacture, a manufacturer, an expiration date, the number of components, and access information (URL or the like) for accessing these pieces of information.

(64) As described above, in the present exemplary embodiment, reading unit **65** is a component information acquisition unit that acquires component information that is information on components stored in carrier tape **27**. Note that, in a case where reel **44** has a radio tag storing component information, a non-contact reader having a function of reading the information of the radio tag in a non-contact manner may be used as reading unit **65**. Reading unit **65** reads the component information from reel **44** of reel supply unit **61**, but may read the component information from reel **44** gripped by gripping head **71**. In the present exemplary embodiment, reading unit **65** is disposed at a position where an identifier of reel **44** of reel supply unit **61** can be read. However, the reading unit may be disposed at any position as long as the identifier can be read. Further, it may be mounted on gripping head **71** or a robot arm (not illustrated) so as to be movable. Further, reading unit **65** may be configured by a wireless or wired hand scanner, and the identifier of reel **44** may be read by an operation of operator OP (see FIG. **14**).

(65) Writing unit **66** writes information in radio tag **42M** attached to case **42** held by case holding part **63Ba** by radio communication. The information written in radio tag **42M** by writing unit **66** is the component information acquired (read) by reading unit **65** and the identification information.

(66) Here, the “identification information” is information used to identify component BH stored in one roll body **27R** used in component mounting system **1** from component BH stored in another roll body **27R**. The identification information includes, for example, a serial number issued in a factory in which component mounting system **1** is installed. In the present exemplary embodiment, as described later, the identification information is generated (issued) by management computer **6** when roll body **27R** separated from reel **44** is stored in case **42**. Further, the identification information is also used as management information for managing case **42**. That is, the identification information is used for both the purpose of identifying roll body **27R** and the purpose of identifying case **42**.

(67) Next, storage warehouse **4** will be described. Storage warehouse **4** stores roll body **41** in a case. Roll body **41** in a case referred to herein includes not only roll body **41** in a case just manufactured by carrier tape processing device **3** but also in-use roll body **41** in a case that has been used in component mounting device **14** and returned halfway.

(68) In FIG. **14**, storage warehouse **4** has a plurality of shelf parts **82** (storage parts) in housing **81**. Inlet **83** is provided below a front surface of housing **81**, and moving table **84** that performs an operation of protruding to the outside of housing **81** (a front side of operator OP) through inlet **83** and an operation of retracting to the inside of housing **81** (a back side as viewed from operator OP) is provided inside housing **81**. Moving table **84** is overhung to the front side of operator OP when roll body **41** in a case is stored in storage warehouse **4**, and is drawn to the back side when roll body **41** in a case placed on moving table **84** is housed in housing **81**.

(69) A plurality of storage positions **82S** for storing (placing) roll body **41** in a case are determined in advance in each of the plurality of shelf parts **82** provided in storage warehouse **4**. That is, in the present exemplary embodiment, each of shelf parts **82** is a storage part including a plurality of

storage positions **82S** for storing case **42** (that is, roll body **41** in a case) storing roll body **27R**.

(70) In FIG. **14**, case transfer mechanism **85** is provided in housing **81**. Case transfer mechanism **85** includes Z-axis table **86** extending in the vertical direction (referred to as the Z-axis direction), Y-axis table **87** extending in the front-rear direction (referred to as the Y-axis direction) as viewed from operator OP and moved in the vertical direction by Z-axis table **86**, X-axis table **88** extending in the lateral direction (referred to as the X-axis direction) as viewed from operator OP and moved in the front-rear direction by Y-axis table **87**, and transfer head **89** moved in the X-axis direction by X-axis table **88**.

(71) Case transfer mechanism **85** moves transfer head **89** three-dimensionally by the movement operation of Y-axis table **87** in the Z-axis direction by Z-axis table **86**, the movement of X-axis table **88** in the Y-axis direction by Y-axis table **87**, and the movement of transfer head **89** in the X-axis direction by X-axis table **88**. Transfer head **89** includes two fingers **89F** arranged in the X-axis direction.

(72) Phase head **89** can bring two fingers **89F** close to or away from each other in the X-axis direction. When two fingers **89F** are actuated to close with roll body **41** in a case positioned between two fingers **89F**, roll body **41** in a case is gripped by two fingers **89F** (i.e., by transfer head **89**).

(73) In FIG. **14**, transfer head **89** is provided with non-contact reader **90** in which an imaging visual field is directed in a depth direction as viewed from operator OP. Non-contact reader **90** faces radio tag **42M** provided in case **42** of roll body **41** in a case at a position where transfer head **89** grips roll body **41** in a case.

(74) Non-contact reader **90** reads information written in radio tag **42M** (component information and identification information of component BH stored in roll body **27R**) by radio communication in a state of facing radio tag **42M**. In the present exemplary embodiment, non-contact reader **90** functions as an identification information reading unit that reads identification information written in radio tag **42M** as a storage unit. When reading the information written in radio tag **42M** of roll body **41** in a case, non-contact reader **90** stores the read information in management computer **6** (described later).

(75) A unique address is given to each of storage positions **82S** defined in each of the plurality of shelf parts **82**. In the present exemplary embodiment, when roll body **41** in a case is stored (placed) in storage position **82S**, information in which information (component information) on component BH stored in roll body **27R** (carrier tape **27**) of roll body **41** in a case is associated (so-called associated) with identification information of roll body **41** in a case is stored in management computer **6**. Further, information in which the component information and the information on storage position **82S** (storage position information) are associated with each other is stored in management computer **6** (described later). As a result, management computer **6** can grasp which storage position **82S** of storage warehouse **4** stores which component BH in which roll body **41** in a case is stored.

(76) FIG. **15** is a block diagram illustrating a control system in entire component mounting system **1**. As illustrated in FIG. **15**, management computer **6** includes production information management unit **91**, component monitoring unit **92**, work instruction unit **93**, and information management unit **94**. Production information management unit **91** stores production plan data in component mounting system **1**. Component monitoring unit **92** monitors the number of remaining components BH in manufacturing line **2**. When a component shortage is predicted, component monitoring unit **92** transmits the fact to work instruction unit **93**.

(77) Work instruction unit **93** gives a work instruction at the time of model switching based on the production plan data stored in production information management unit **91** to manufacturing line **2** and operator OP, and gives an instruction such as delivery of component BH to be used in production of the next model to storage warehouse **4**. Work instruction unit **93** issues a work instruction (specifically, an instruction to storage warehouse **4** to dispense the component for

replenishment and an instruction to operator OP to perform the component replenishment work) based on the information from component monitoring unit **92**.

(78) Information management unit **94** includes first information management unit **94a**, second information management unit **94b**, identification information generation unit **94c**, and storage position information generation unit **94d** (FIG. **15**). First information management unit **94a** manages the component information, and second information management unit **94b** manages information (storage position information) of storage position **82S** of component BH in storage warehouse **4**. Specifically, first information management unit **94a** stores the component information and the identification information of roll body **27R** separated from reel **44** in association with each other. Further, second information management unit **94b** stores the identification information on roll body **27R** separated from reel **44** and the information on storage position **82S** (storage position information) in association with each other.

(79) Identification information generation unit **94c** generates (issues) the identification information on roll body **27R** when carrier tape processing device **3** disassembles reel **44** of roll body **43** with a reel and roll body **27R** is separated from reel **44**. When roll body **41** in a case is stored in storage warehouse **4**, storage position information generation unit **94d** specifies storage position **82S** of roll body **41** in a case and generates the storage position information.

(80) As illustrated in FIG. **15**, carrier tape processing device **3** includes processing device controller **3C**. Processing device controller **3C** controls reel conveyance unit **62**, roll body taking-out unit **63**, reading unit **65**, writing unit **66**, and the like included in carrier tape processing device **3**. As illustrated in FIG. **15**, storage warehouse **4** includes warehouse controller **4C**. Warehouse controller **4C** controls moving table **84**, case transfer mechanism **85**, and non-contact reader **90** included in storage warehouse **4**.

(81) Next, with reference to a flowchart illustrated in FIG. **16**, a flow of a manufacturing operation (a processing operation of carrier tape **27**) of roll body **41** in a case by carrier tape processing device **3** will be described. First, processing device controller **3C** of carrier tape processing device **3** acquires component information of roll body **43** with a reel to be worked out of roll body **43** with a reel supplied by reel supply unit **61** (component information acquiring step in step ST1). Here, the acquisition of the component information is performed by reading unit **65** reading the identifier of code label **43L** attached to reel **44** of roll body **43** with a reel to be worked (FIG. **9**), but another method may be used. For example, operator OP may perform an input corresponding to the content of code label **43L** from an input device (not illustrated) connected to processing device controller **3C**.

(82) When the component information of roll body **43** with a reel is acquired in step ST1, processing device controller **3C** operates reel conveyance unit **62** of carrier tape processing device **3** to cause gripping head **71** to grip roll body **43** with a reel to be worked. Then, when gripping head **71** grips roll body **43** with a reel, gripping head **71** is moved (FIG. **10**) to position roll body **43** with a reel above separator **73** (positioning step in step ST2. FIG. **11**).

(83) After positioning roll body **43** with a reel above separator **73**, processing device controller **3C** disassembles reel **44** by separator **73** as described above (FIG. **12A**.fwdarw.FIG. **12B**.fwdarw.FIG. **12C**), and separates roll body **27R** from reel **44** (separation step in step ST3). As a result, reel **44** is divided into two parts, that is, a part including one of side plates **45** and a part including the other of side plates **45**, and roll body **27R** passes through roll body storage unit **63B** and is stored in empty case **42** located immediately below roll body storage unit **63B** (roll body storing step in step ST4). Each of two divided reels **44** passes through reel discharge chute **75** and is recovered in recovery box **76** (holding body recovery step in step ST5).

(84) Here, the separation step in step ST3 and the roll body storing step in step ST4 are a roll body taking-out and storing step in which roll body **27R** is taken out from reel **44** and stored in case **42**. Here, reel **44** is a holding body that holds carrier tape **27** storing component BH in a state of roll body **27R**. In the roll body taking-out and storing step, in the roll body storing step, roll bodies **27R**

separated from reel **44** in the separation step are stored one by one in empty case **42**. Therefore, one roll body **27R** can be managed in units of one case **42**.

(85) When separating reel **44** in step ST3, processing device controller **3C** requests identification information generation unit **94c** of management computer **6** to generate the identification information on roll body **27R** separated from reel **44**. Then, identification information generation unit **94c** requested by processing device controller **3C** to generate the identification information generates the identification information on roll body **27R** taken out by separation of reel **44** (identification information generation process of step ST6).

(86) When the identification information is generated by identification information generation unit **94c** in step ST6, processing device controller **3C** of carrier tape processing device **3** receives the generated identification information. Then, writing unit **66** writes the component information acquired in step ST1 and the identification information generated in step ST6 in radio tag **42M** attached to case **42** (case **42** in which roll body **27R** is stored in step ST4) (writing step in step ST7).

(87) After writing the component information and the identification information in radio tag **42M** of case **42**, processing device controller **3C** transmits the information written in radio tag **42M** (the component information and the identification information on roll body **27R**) to management computer **6**, and causes first information management unit **94a** of management computer **6** to store the information (the component information and the identification information). As a result, for roll body **27R** separated from reel **44**, the identification information, which is the information unique to roll body **27R**, and the component information, which is the information on component BH stored in roll body **27R**, are registered in management computer **6** in association with each other (component information registration step in step ST8).

(88) After writing the information on roll body **27R** in first information management unit **94a** of management computer **6**, processing device controller **3C** ends the manufacture of roll body **41** in a case by carrier tape processing device **3** (processing operation of carrier tape **27**).

(89) Next, a flow of an operation of storing roll body **41** in a case will be described with reference to a flowchart illustrated in FIG. 17. First, warehouse controller **4C** of storage warehouse **4** moves moving table **84** to a front side from inlet **83** provided in housing **81**. Then, operator OP (or a mobile robot (not illustrated)) places roll body **41** in a case on moving table **84** (placing step in step ST11, FIG. 14). At this time, roll body **41** in a case is placed on moving table **84** in an attitude in which upper opening **55** faces upward and front wall **53** faces the front side (the side of operator OP). As a result, radio tag **42M** provided in case **42** faces the front side (FIG. 14).

(90) When roll body **41** in a case is placed on moving table **84**, warehouse controller **4C** moves moving table **84** to draw roll body **41** in a case into housing **81**. When roll body **41** in a case is drawn into housing **81**, warehouse controller **4C** moves transfer head **89** to the front side of roll body **41** in a case placed on moving table **84**. Then, two fingers **89F** are operated to close to cause transfer head **89** to grip roll body **41** in a case. At this time, non-contact reader **90** provided in transfer head **89** faces radio tag **42M** provided in case **42** of roll body **41** in a case, and non-contact reader **90** reads the identification information written in radio tag **42M** (identification information reading step in step ST12).

(91) When non-contact reader **90** reads the identification information written in radio tag **42M**, warehouse controller **4C** transmits the identification information read by non-contact reader **90** to management computer **6**. Then, management computer **6** having received the transmission of the identification information from warehouse controller **4C** specifies storage position **82S** in storage warehouse **4** of roll body **41** in a case corresponding to the identification information (roll body **41** in a case from which non-contact reader **90** has read the identification information) in storage position information generation unit **94d**, and generates the storage position information (storage position information generation step in step ST13). Storage position **82S** is specified by being arbitrarily selected from storage positions **82S** that are in an empty state at that time or according to

a predetermined rule.

(92) When storage position information generation unit **94d** generates the storage position information in step ST13, management computer **6** causes second information management unit **94b** to store the generated storage position information and the identification information of roll body **41** in a case corresponding to the storage position information in association with each other. As a result, for roll body **41** in a case to be stored in storage warehouse **4**, the identification information, which is unique information of roll body **27R**, and the information of the storage position (storage position information) of roll body **41** in a case including roll body **27R** in storage warehouse **4** are registered in management computer **6** in association with each other (storage position information registration step in step ST14).

(93) When management computer **6** causes second information management unit **94b** to store the storage information and the identification information on roll body **41** in a case placed on moving table **84** in association with each other, management computer **6** operates case transfer mechanism **85** to transfer roll body **41** in a case placed on moving table **84** to storage position **82S** corresponding to the storage position information to store (storage step in step ST15). Accordingly, the storage operation of roll body **41** in a case is completed.

(94) As described above, in the present exemplary embodiment, roll body **27R** can be taken out from a holding body such as reel **44** and stored in case **42**, and roll body **27R** can be handled as roll body **41** in a case. Roll body **41** in a case can be made compact by making the size in the width direction smaller than that of roll body **43** with a reel, and reel **44** does not remain as a waste after roll body **27R** is used up (that is, carrier tape **27**), so that workability is good. Further, case **42** itself is inexpensive, and can be reused (reused) differently from reel **44**, so that the cost can be reduced also in this respect.

(95) Furthermore, in the present exemplary embodiment, radio tag **42M** as a storage unit capable of storing component information is provided in case **42**, and the component information recorded on reel **44** can be written and stored in radio tag **42M**. Therefore, even when roll body **27R** is taken out from reel **44**, the component information is not lost, and the component information can be managed in each unit of roll body **41** in a case. In addition, since the component information and the identification information, which is information unique to roll body **27R**, are stored in association with each other, the component information can be managed in each unit of roll body **41** in a case.

(96) In component mounting system **1** described above, carrier tape processing device **3** (processing device controller **3C**, reel conveyance unit **62**, roll body taking-out unit **63**, reading unit **65** as component information acquisition unit, and writing unit **66**), storage warehouse **4** (warehousing controller **4C**, shelf parts **82**, case transfer mechanism **85**, and non-contact reader **90** as identification information reading unit), and information management unit **94** (first information management unit **94a**, second information management unit **94b**, identification information generation unit **94c**, and storage position information generation unit **94d**) constitute component management device **100** (FIG. 15).

(97) In the management work (component management method) of component BH using component management device **100**, first, the component information is acquired (component information acquisition step), the identification information is generated (identification information generation step), the component information and the identification information are written in radio tag **42M** provided in case **42** (writing step), and the written component information and the identification information are associated with each other and stored in first information management unit **94a** (component information registration step). The identification information reading unit (non-contact reader **90**) reads the identification information written in radio tag **42M** (identification information reading step), the storage position information generation unit generates the storage position information by specifying storage position **82S** (storage position information generation step), and second information management unit **94b** stores the identification information

and the storage position information in association with each other (storage position information registration step).

(98) According to such a component management method, since the component information and the identification information of roll body **41** in a case are stored in association with each other, and the identification information and the storage position information of roll body **41** in a case are stored in association with each other, it is possible to smoothly store and take out roll body **41** in a case in storage warehouse **4**. Therefore, according to component management device **100** (component management method) in the present exemplary embodiment, it is possible to easily and efficiently store and unload components BH.

(99) Note that roll body **27R** of roll body **41** in a case handled by component management device **100** in the present exemplary embodiment is not necessarily taken out from roll body **43** with a reel, and may be manufactured as roll body **27R** of carrier tape **27** from the beginning. For example, roll body **27R** may be manufactured by winding a large capacity carrier tape from a roll body while measuring a length of the carrier tape, which is a basis for manufacturing the roll body **43** with a reel.

(100) FIG. **18** illustrates an image of an operation of roll body **41** in a case in component mounting system **1** in the present exemplary embodiment. As illustrated in this drawing, roll body **41** in a case in the present exemplary embodiment is manufactured by supplying roll body **43** with a reel and case **42** to carrier tape processing device **3**. A part of roll body **41** in a case manufactured by carrier tape processing device **3** is sent to manufacturing line **2** to be used, and the other part is stored (stored) in storage warehouse **4**. Roll body **41** in a case stored in storage warehouse **4** is delivered therefrom and sent to manufacturing line **2** to be used. Case **42** caused by carrier tape **27** of roll body **41** in a case being used up in manufacturing line **2** is recovered and supplied to carrier tape processing device **3** to be reused for manufacturing of new roll body **41** in a case. A part of roll body **41** in a case sent to manufacturing line **2** in a state where carrier tape **27** has not been used up is returned to storage warehouse **4** and stored (received).

(101) As described above, in carrier tape processing device **3** (carrier tape processing method) according to the present exemplary embodiment, roll body **27R** can be taken out from the holding body such as reel **44** and stored in case **42**, and roll body **27R** can be handled as roll body **41** in a case. Roll body **41** in a case can be made compact by making the size in the width direction smaller than that of roll body **43** with a reel, and reel **44** does not remain as a waste after roll body **27R** is used up (that is, carrier tape **27**), so that workability is good.

(102) Further, case **42** itself is inexpensive, and can be reused (reused) differently from reel **44**, so that the cost can be reduced also in this respect. Therefore, according to carrier tape processing device **3** (carrier tape processing method) of the present exemplary embodiment, a component supply form without a reel can be realized with an inexpensive configuration.

(103) Furthermore, in the present exemplary embodiment, radio tag **42M** as a storage unit capable of storing component information is provided in case **42**, and the component information recorded on reel **44** can be written and stored in radio tag **42M**. Therefore, even when roll body **27R** is taken out from reel **44**, the component information is not lost, and the component information can be managed in units of each roll body **41** in a case. In addition, since the component information and the identification information, which is information unique to roll body **27R**, are stored in association with each other, the component information can be managed in each unit of roll body **41** in a case. Therefore, according to carrier tape processing device **3** (carrier tape processing method) of the present exemplary embodiment, it is possible to realize a component supply form without a reel with an inexpensive configuration while managing component information.

(104) While the exemplary embodiment of the present disclosure has been described above, the present disclosure is not limited to the above exemplary embodiment, and various modifications and the like can be made. For example, in the above-described exemplary embodiment, reel **44** of roll body **43** with a reel is divided into two. However, this is an example, and the reel may be

divided into three or more parts. Even in this case, it is sufficient that each of the plurality of parts generated by the disassembly passes through reel discharge chute **75** and falls into recovery box **76**, whereby disassembled reel **44** can be reliably recovered.

(105) Further, the configuration of the disassembly unit of reel **44** and the disassembly method described in the above-described exemplary embodiment are merely examples, and roll body **27R** may be taken out from reel **44** by disassembling with another method. Furthermore, as long as reel **44** can be removed without disassembling reel **44** (for example, a configuration in which one side plate **45** of reel **44** is provided, winding core **46** extends from one side plate **45** in a cantilevered state, and roll body **27R** can be separated from reel **44** when roll body **27R** is removed from winding core **46**), it is not always necessary to disassemble reel **44**. Furthermore, the specific configuration of roll body taking-out unit **63** described in the above-described exemplary embodiment is merely an example, and the configuration is arbitrarily as long as roll body **27R** can be taken out from the holding body such as reel **44** that holds carrier tape **27** storing component BH in the state of roll body **27R** and stored in case **42**.

INDUSTRIAL APPLICABILITY

(106) To provide a carrier tape processing device and a carrier tape processing method capable of realizing a component supply form without a reel with an inexpensive configuration.

REFERENCE MARKS IN THE DRAWINGS

(107) **3**: carrier tape processing device **27**: carrier tape **27R**: roll body **41**: roll body in a case **42**: case **42M**: radio tag (storage unit) **43**: roll body with a reel **44**: reel (holding body) **45**: side plate **46**: winding core **63**: roll body taking-out unit **63A**: separation unit **63B**: roll body storage unit **63Ba**: case holding part **63Bb**: chute part **64**: recovery unit **65**: reading unit (component information acquisition unit) **66**: writing unit **82**: shelf part (storage part) **94**: information management unit **94a**: first information management unit **94b**: second information management unit **94c**: identification information generation unit **94d**: storage position information generation unit BH: component

Claims

1. A carrier tape processing device comprising a roll body taking-out unit that takes out a roll body from a holding body and stores the roll body in a case, the holding body being a reel and holding a carrier tape storing components in a state of the roll body, wherein the roll body taking-out unit includes: a separation unit that separates the roll body from the holding body; and a roll body storage unit that stores, into the case, the roll body separated from the holding body by the separation unit.
2. The carrier tape processing device according to claim 1, wherein the roll body storage unit includes: a case holding part that holds the case; and a chute part that stores the roll body separated from the holding body by the separation unit one by one in the case held by the case holding part.
3. The carrier tape processing device according to claim 1, wherein the separation unit separates the roll body from the holding body by disassembling the holding body.
4. The carrier tape processing device according to claim 1, further comprising a recovery unit that recovers the holding body from which the roll body has been separated by the separation unit.
5. The carrier tape processing device according to claim 1, wherein the reel includes a winding core around which the carrier tape is wound and a pair of side plates sandwiching the winding core, and the separation unit disassembles the holding body by dividing the holding body into two parts, a part including one of the side plates and a part including another of the side plates.
6. A carrier tape processing method comprising a roll body taking-out and storing step of taking out a roll body from a holding body that is a reel and holds a carrier tape storing components in a state of the roll body, and storing the roll body in a case, wherein the roll body taking-out and storing step includes: a separation step of separating the roll body from the holding body; and a roll body

storing step of storing the roll body separated from the holding body in the separation step in the case.

7. The carrier tape processing method according to claim 6, wherein in the roll body storing step, the roll body separated from the holding body in the separation step is stored one by one in the case.

8. The carrier tape processing method according to claim 6, wherein in the separation step, the roll body is separated from the holding body by disassembling the holding body.

9. The carrier tape processing method according to claim 6, wherein the reel includes a winding core around which the carrier tape is wound and a pair of side plates sandwiching the winding core, and in the separation step, the holding body is divided into two parts, a part including one of the side plates and a part including another of the side plates, and disassembled.

10. The carrier tape processing method according to claim 6, further comprising a holding body recovering step of recovering the holding body from which the roll body has been separated in the separation step.

11. A carrier tape processing device comprising: a roll body taking-out unit that takes out a roll body from a holding body that is a reel and holds a carrier tape storing components in a state of the roll body, and stores the roll body in a case including a storage unit; a component information acquisition unit that acquires component information that is information on the components; and a writing unit that writes the component information acquired by the component information acquisition unit in the storage unit, wherein the roll body taking-out unit includes: a separation unit that separates the roll body from the holding body; and a roll body storage unit that stores, into the case, the roll body separated from the holding body by the separation unit.

12. The carrier tape processing device according to claim 11, wherein the component information is recorded in the holding body, and the component information acquisition unit acquires the component information by reading the component information recorded in the holding body.

13. The carrier tape processing device according to claim 12, wherein the component information acquisition unit reads the component information before the roll body is separated from the holding body.

14. The carrier tape processing device according to claim 11, further comprising an identification information generation unit that generates identification information used to identify the components from a component stored in another case, wherein the writing unit writes the identification information together with the component information.

15. The carrier tape processing device according to claim 14, further comprising an information management unit that stores the component information read by the component information acquisition unit and the identification information generated by the identification information generation unit in association with each other.

16. The carrier tape processing device according to claim 11, wherein the component information includes at least one of a component type, a component name, a characteristic, a date of manufacture, a manufacturer, an expiration date, a number of components, and access information for accessing these pieces of information.

17. A carrier tape processing method comprising: a roll body taking-out and storing step of taking out a roll body from a holding body that is a reel and holds a carrier tape storing components in a state of the roll body, and storing the roll body in a case including a storage unit; and a writing step of writing component information that is information on the components, in the storage unit, wherein the roll body taking-out and storing step includes: a separation step of separating the roll body from the holding body; and a roll body storing step of storing the roll body separated from the holding body in the separation step in the case.

18. The carrier tape processing method according to claim 17, further comprising a component information acquisition step of acquiring the component information recorded in the holding body, wherein, in the writing step, the component information acquired in the component information

acquisition step is written in the storage unit.

19. The carrier tape processing method according to claim 18, wherein in the component information acquisition step, the component information is read and acquired before the roll body is separated from the holding body.

20. The carrier tape processing method according to claim 18, further comprising an identification information generation step of generating identification information used to identify the components from a component stored in another case, wherein in the writing step, the identification information is written together with the component information.

21. The carrier tape processing method according to claim 20, further comprising a component information registration step of storing the component information acquired in the component information acquisition step and the identification information generated in the identification information generation step in association with each other.

22. The carrier tape processing method according to claim 17, wherein the component information includes at least one of a component type, a component name, characteristics, a date of manufacture, a manufacturer, an expiration date, a number of components, and access information for accessing these pieces of information.
